# Analysis of Multi-agent performative prediction in "Bank-games"

Project report for ECE-8803 Online Decision Making in Machine Learning, by Srikanth Avasarala and Karthic Palaniappan

## 1 Motivation

Traditional machine learning models are often developed under the assumption that data is static and independently and identically distributed (i.i.d.), with no consideration of how model predictions might influence future data distributions. However, in many high-stakes applications such as lending, hiring, and healthcare, this assumption breaks down. Instead, predictions and decisions actively shape future behavior and data, leading to feedback loops that alter the data distribution over time. This dynamic relationship between models and their environments is referred to as performative prediction.

Moreover, most existing research in performative prediction focuses on single-agent settings, ignoring how competition between multiple decision-makers (e.g., banks, schools, or insurance companies) further compounds these dynamics. In practice, organizations often compete for users or customers, and their decisions influence not only the outcomes of their own models but also the data available to their competitors. Self-selection effects—where individuals choose which institution to engage with based on the institutions' policies—can create additional biases and strategic interactions that are not well understood.

To understand some progress towards addressing this gap, we study and replicate a recently submitted paper [3] on this area. This paper introduces a stylized competitive setting between two banks—dubbed the Bank Game—to explore how strategic interaction and self-selection influence performative prediction dynamics. The model sheds light on the behavior of simple learning algorithms (like exponential weights) in settings that violate standard assumptions like insensitivity and static distributions. Surprisingly, despite these complexities, the authors show that these learning dynamics can converge to stable equilibria under realistic conditions.

Our goal in replicating this study is to empirically validate the theoretical findings and better understand how performative feedback and strategic competition influence model learning in dynamic environments. Given the amount of overlap this area of research has with course contents of ECE-8803 including topics like - general sum game theory, adversarial prediction and online learning (Multiplicative Weights algorithm has been used in this paper) we found this an interesting piece of literature to study as part of our project. Our experimental results are available at https://github.com/i-like-bfs-and-dfs/multi-agent-performative-prediction - they include convergence analysis with full information of the customer distribution, as well as from limited information on distribution (i.e., sampling from distribution).

## 2 Introduction

Bertrand-style competition is a model in economics where firms compete to offer lower prices for homogeneous products to capture customer demand. [3] discusses a 2-player general sum Bank Game, where each bank competes to offer lower interest rate and credit score threshold to capture customer demand distribution and optimize expected return. Insensitivity assumptions do not hold here, since a marginally smaller interest rate from one bank leads to that bank capturing the entire demand and resulting in a drastic shift in utility.

A one-shot analysis of this game is hopeless [1], since Bertrand competitions suffer from Prisoner's dilemma where each entity has no gain when changing strategies, leading to both entities selecting a bad strategy. No-regret dynamics cannot be guaranteed in this case. Due to the general-sum nature, the problem cannot be solved using simple linear programming techniques.

[3] presents a multiple-round performative prediction model based on Exponential Weights modified to guide convergence to optimal (Nash) equilibrium. Due to adversarial opponents, the framework induces

collaborative selection of strategies to optimize utility. Keeping in mind the above mentioned considerations, the authors propose a dynamic performative Bank game with no assumption on insensitivity of parameters, (it in fact is agnostic to any sensitivity) which we discuss through out the report.

# 3 Model Overview

In this section, we outline the model used in the paper. The game involves a *duopoly* of mortgage companies, or banks, competing to sell loans to customers.

## 3.1 Customer Model

To illustrate the equilibria and dynamics of performative prediction games, we focus on a scenario where two competing mortgage companies, or banks, aim to attract customers by offering loans. The population of potential customers is denoted as $\mathcal{P}$, where each individual in this population has a one-dimensional type, represented by $y_j \in [0, 1]$. For simplicity, we assume that $y_j$ reflects the customer's probability of repaying the loan, i.e., $y_j := \mathbb{P}[Y_j = 1]$, where $Y_j$ is a random variable indicating whether the customer repays the loan ($Y_j = 1$) or defaults on it ($Y_j = 0$). The types of customers in the population are drawn from a known distribution $D_y$ supported on the interval $[0, 1]$.

In the game between the two banks, each bank $i \in \{1, 2\}$ selects a pair of parameters $(\tau_i, \gamma_i) := \theta_i$, where $\tau_i$ is the credit score threshold and $\gamma_i$ is the interest rate. Specifically, $\tau_i$ can take one of two values, $\tau_l$ or $\tau_h$, representing the credit score threshold for approving a customer. A customer $j$ is approved by bank $i$ if and only if their credit score $y_j$ is greater than or equal to the threshold $\tau_i$. The interest rate $\gamma_i$ is chosen from two possible values, $\gamma_l$ and $\gamma_h$, and it determines the rate offered to approved customers. The allowable values for the thresholds are denoted by the set $\Gamma := [0, 1]$, and the interest rates are denoted by $\Lambda := [0, 1]$.

We normalize the loan amount to 1, without loss of generality, so that the expected utility for a bank that approves a customer is given by $(1 + \gamma_i) \cdot \mathbb{P}[Y_i = 1] - \mathbb{P}[Y_i = 0] = (1 + \gamma_i)y_i - (1 - y_i)$.

Each customer makes a decision based on the parameters selected by both banks. If a customer's credit score $y$ qualifies them for only one bank's loan (i.e., their score is above the threshold for one bank but not the other), they choose that bank. If a customer qualifies for both banks, they select the bank offering the lower interest rate. If both banks offer the same rate, the customer chooses randomly between them. Finally, if a customer is unqualified for both banks (i.e., their credit score is below both banks' thresholds), they are rejected by both.

The expected reward for Bank 1, given the parameters $\theta_1 = (\tau_1, \gamma_1)$ and $\theta_2 = (\tau_2, \gamma_2)$ set by both banks, can be expressed as:

$$
\begin{aligned}
u_1(\theta_1, \theta_2) = \mathbb{E}_{y \sim D_y} &\left[ \mathbb{I}\{\underbrace{\tau_1 \leq y < \tau_2 \ \cup \ (\tau_1, \tau_2 \leq y \ \cap \ \gamma_1 < \gamma_2)}_{\text{accepted by Bank 1}}\} \cdot \big((1 + \gamma_1)y - (1 - y)\big) \right] \\
&+ \frac{1}{2}\mathbb{E}_{y \sim D_y} \left[ \mathbb{I}\{\underbrace{\tau_1, \tau_2 \leq y \ \cap \ \gamma_1 = \gamma_2}_{\text{accepted by both Banks}}\} \cdot \big((1 + \gamma_1)y - (1 - y)\big) \right]. \qquad (1)
\end{aligned}
$$

This utility function is symmetric, meaning that the utility for Bank 2 can be derived by swapping the roles of $\theta_1$ and $\theta_2$. Specifically, $u_2(\theta_1, \theta_2) = u_1(\theta_2, \theta_1)$, reflecting the game's symmetry.

In summary, this setup models a duopoly between two mortgage banks, where each bank strategically chooses a credit score threshold and interest rate, and customers decide which bank to approach based on these parameters. The expected utility for each bank is influenced by the distribution of customer types and the choices made by the other bank, leading to a dynamic game where equilibrium analysis can reveal stable outcomes under various assumptions about customer behavior and parameter settings.

### 3.2 Equilibrium Concepts

We start by defining the main solution concepts used in this section, in the *one-shot* setting. The main solution concept used in this paper is the concept of *Nash equilibrium*:

**Definition 1** (Nash equilibrium)**.** *A pair of strategies* $(\theta_1^*, \theta_2^*)$ *constitutes a* pure Nash equilibrium *(pure NE) if and only if:*

$$u_1(\theta_1^*, \theta_2^*) \geq \max_{\theta_1 \in \Gamma \times \Lambda} u_1(\theta_1, \theta_2^*), \quad \textit{and} \quad u_2(\theta_1^*, \theta_2^*) \geq \max_{\theta_2 \in \Gamma \times \Lambda} u_2(\theta_1^*, \theta_2).$$

**Definition 2.** *A pair of mixed strategies* $(p_1^*, p_2^*)$ *constitutes a* mixed Nash equilibrium *(mixed NE) if and only if the following conditions hold:*

$$\mathbb{E}_{\theta_1 \sim p_1^*, \theta_2 \sim p_2^*}[u_1(\theta_1, \theta_2)] \geq \max_{p \in \Delta(\Gamma \times \Lambda)} \mathbb{E}_{\theta_1 \sim p, \theta_2 \sim p_2^*}[u_1(\theta_1, \theta_2)],$$

*and*

$$\mathbb{E}_{\theta_1 \sim p_1^*, \theta_2 \sim p_2^*}[u_2(\theta_1, \theta_2)] \geq \max_{p \in \Delta(\Gamma \times \Lambda)} \mathbb{E}_{\theta_1 \sim p_1^*, \theta_2 \sim p}[u_2(\theta_1, \theta_2)].$$

In Nash equilibria, players pick their actions simultaneously and independently of each other. Beyond Nash Equilibria, th authors's introduce another concept of Correlated Equilibria (CEs) of a game, where coordination among players is allowed: we do not include this part in the report for analysis/implementation.

Informally, correlated equilibria use a correlation or signaling device that recommends an action $(\theta_1, \theta_2)$ to the players according to a *joint* distribution (e.g., a traffic signal at an intersection). At a correlated equilibrium, no player benefits from deviating from the recommended action in expectation.

### 3.3 One-Shot Equilibrium Characterization for $n = 2$ Actions

To simplify presentation, we introduce the following function, which is useful in writing down players' utilities:

$$f_{D_y}(\gamma, \tau_a, \tau_b) \triangleq \int_{\tau_a}^{\tau_b} [(2+\gamma)y - 1]p(y)\, dy. \tag{2}$$

We often omit the subscript $D_y$ when it is clear from context. Intuitively, this represents the utility for a single bank that sets the interest rate as $\gamma$ and is able to attract all the customers in the range $Y \in [\tau_a, \tau_b]$.

As discussed in Section 3.1, we focus on the simplest case where there are two choices for each parameter: $0 \leq \tau_\ell < \tau_h \leq 1$ and $0 \leq \gamma_\ell < \gamma_h \leq 1$. In this scenario, the utility of Bank 1 under each pair of decisions forms a $4 \times 4$ matrix (as shown in Table 1). We consider the canonical case where:

$$\tau_\ell = \frac{1}{2+\gamma_h}, \quad \text{and} \quad \tau_h = \frac{1}{2+\gamma_\ell}.$$

These choices for the thresholds are natural from Equation (2). Indeed, it is a dominant strategy for a *rational* bank with interest rate $\gamma$ to set a threshold of $\tau^*(\gamma) := \frac{1}{2+\gamma}$. Setting a higher threshold $\tau > \tau^*(\gamma)$ leads to ignoring customers in the range $y \in [\tau^*(\gamma), \tau]$ that provide utility $(2+\gamma)y - 1 > 0$, while setting a lower threshold $\tau < \tau^*(\gamma)$ risks attracting customers with $y < \tau^*(\gamma)$, which yields negative utility. Both cases lead to a utility loss.

### 3.4 Model dynamics

In the last sub-section we discussed one-shot equilibrium i.e the case where each agent makes a choice with respect to changing/keeping parameters. In real world settings, the interaction between bank and agents in population occur for a long enough time for a considerable part of the population to have played the game. In order to model the fynamics, the paper uses Exponential Weights which is formally presented in Algorithm 1. At each time step $t$, each player $i \in \{1, 2\}$ selects an action $\theta^{(j)}$ according to their current probability distribution $p_i^{(t)}$. After this decision, both players update their distributions for the next round. The update for each action is proportional to the exponential of the expected utility that action yields,

| | $(\tau_\ell, \gamma_\ell)$ | $(\tau_\ell, \gamma_h)$ | $(\tau_h, \gamma_\ell)$ | $(\tau_h, \gamma_h)$ |
|---|---|---|---|---|
| $(\tau_\ell, \gamma_\ell)$ | $\frac{1}{2}f(\gamma_\ell, \tau_\ell, 1)$ | $0$ | $\frac{1}{2}f(\gamma_\ell, \tau_h, 1)$ | $0$ |
| $(\tau_\ell, \gamma_h)$ | $f(\gamma_\ell, \tau_\ell, 1)$ | $\frac{1}{2}f(\gamma_h, \tau_\ell, 1)$ | $f(\gamma_\ell, \tau_h, 1)$ | $\frac{1}{2}f(\gamma_h, \tau_h, 1)$ |
| $(\tau_h, \gamma_\ell)$ | $f(\gamma_\ell, \tau_\ell, \tau_h) + \frac{1}{2}f(\gamma_\ell, \tau_h, 1)$ | $f(\gamma_h, \tau_\ell, \tau_h)$ | $\frac{1}{2}f(\gamma_\ell, \tau_h, 1)$ | $0$ |
| $(\tau_h, \gamma_h)$ | $f(\gamma_\ell, \tau_\ell, 1)$ | $f(\gamma_h, \tau_\ell, \tau_h) + \frac{1}{2}f(\gamma_h, \tau_h, 1)$ | $f(\gamma_\ell, \tau_h, 1)$ | $\frac{1}{2}f(\gamma_h, \tau_h, 1)$ |

Table 1: Utility matrix for Bank 1. Rows denote strategies of Bank 2; columns denote strategies of Bank 1.

given the opponent's current mixed strategy. For instance, in round $t$, the expected utility of action $\theta^{(j)}$ for player 1—assuming player 2 plays according to $p_2^{(t)}$—is $\sum_{k=1}^{4} p_{2,k}^{(t)} \cdot u_1\left(\theta^{(j)}, \theta^{(k)}\right)$.

A crucial parameter in the algorithm is the learning rate $\eta$. To achieve standard no-regret guarantees, $\eta$ is typically set to a time-decaying value like $\eta = \Theta(1/\sqrt{T})$ (for a known time horizon $T$) or $\eta_t = \Theta(1/\sqrt{t})$. Remarkably, however, the theoretical properties we analyze continue to hold even for any fixed step size $\eta \in (0, \infty)$. An important point to note is that the utility function computed for the exponential

**Algorithm 1** Online Learning Dynamic via Exponential Weights

**Input:** step size $\eta > 0$, time horizon $T$
**Initialize:** $p_1^{(1)}, p_2^{(1)} \in \Delta_4$
**for** $t = 1$ to $T$ **do**
  *Decision:* Each player $i \in \{1, 2\}$ plays action $\theta^{(j)}$ with probability $p_{i,j}^{(t)}$
  *Update:* For each $j \in [4]$, update probabilities as follows:
    $p_{1,j}^{(t+1)} \propto p_{1,j}^{(t)} \cdot \exp\left(\eta \sum_{k=1}^{4} p_{2,k}^{(t)} \cdot \hat{u}_1^{(t)}(\theta^{(j)}, \theta^{(k)})\right)$
    $p_{2,j}^{(t+1)} \propto p_{2,j}^{(t)} \cdot \exp\left(\eta \sum_{k=1}^{4} p_{1,k}^{(t)} \cdot \hat{u}_2^{(t)}(\theta^{(k)}, \theta^{(j)})\right)$
**end for**

weights is computed assuming the customer distribution is known. One of the novel elements of this paper is the introduction of case where population distribution is unknown and we have access to samples at every time step. The utility function for each bank at every time step $\hat{u}_j^{(t)}$ $j \in \{1, 2\}$ is then computed in a Monte-Carlo fashion using the samples from customer distribution.

## 4 Theoretical convergence result

In this section, we present one theoretical result from the paper and state some main ideas behind the working of the proof. The result shows the convergence to Nash equilibrium of dynamics via the Exponential Weights algorithm discussed in the last section for a certain initial probabilities of parametetric choices.

**Case I:** $\epsilon_1 < 0, \epsilon_2 < 0$**:** Then, Algorithm 1 approximately converges to $((\tau_\ell, \gamma_h), (\tau_\ell, \gamma_h))$. Specifically:

**Theorem 1** (Part I)**.** *Suppose $\epsilon_1 < 0, \epsilon_2 < 0$. Let $t_s'' = t_s' + 1 + \left\lceil \frac{4}{\eta\xi_1} \log \frac{4(\xi_1+3)c_{ini}}{\xi_1} \right\rceil$ be a constant. For Algorithm 1, for any error tolerance $\omega > 0$ and $\delta \in (0, 1)$, if $T \geq \frac{4}{\xi_1}\left(3 + \frac{\xi_1}{4}\right) t_s'' + \frac{4}{\eta\xi_1} \log \frac{c_{ini}}{\omega}$, and $n_{samples} \geq \frac{1280 \log \frac{64T}{\delta}}{\xi_1^2}$, then with probability at least $1 - \delta$, for $i \in [2]$, we have*

$$p_{i,2}^{(T)} \leq \omega, \quad \textit{and} \quad p_{i,3}^{(T)} \geq 1 - \min\{3\omega, 1\}.$$

Let's breakdown the result in simple words. The above theorem states that for a small error tolerance $\omega > 0$, and with a number of samples $n_{\text{samples}}$ that scales inversely with $\omega^2$, we can estimate the expected utilities accurately enough using Monte Carlo sampling. Then, for a sufficiently large time

horizon $T$—which depends logarithmically on the initial conditions and inversely on the payoff gap parameter $\xi_1$—the exponential weights update ensures that the probability of playing suboptimal strategy $\theta^{(2)}$ decreases to at most $\omega$, while the probability of playing the optimal strategy $\theta^{(3)}$ increases to at least $1 - \min\{3\omega, 1\}$ with high probability.Therefore the convergence is towards pure nash equilibrium as $p_{i,2}^{(T)} \longrightarrow 0$, and $p_{i,3}^{(T)} \longrightarrow 1$, and the probabilities for other parameteric choices ($p_{i,2}^{(T)} = 0, p_{i,2}^{(T)} = 0$. The proof strategy involves using Hoeffding's inequality to bound finite sample computation of utility function, and first principle arguments for convergence involving exponential functions. An important point to note is that this proof does not use any information related to regret bound of exponential Weights algorithm.

# 5 Experiments

We replicated the experiments in Table 2 and observed similar convergences for all 6 cases. The utility function is available on the Git repo with parameters and distribution (truncated gaussian/piecewise uniform) as arguments. Exponential weights function implementation takes in the utility matrix as argument to generate probabilities for T timesteps with $\eta$ learning rate.

```
python multi-agent-performative-prediction.py -d gauss
-m 0.1 -s 0.3 -gl 0.4 -gh 0.8 -T 50000
```

Figure 1: Sample command to run exponential weights algorithm with truncated Gaussian (full) distribution.

| **Distribution** | **Equilibria** | **Dynamics of Algorithm** |
|---|---|---|
| $D_1(0.1, 0.3)$ | $((\tau_l, \gamma_h), (\tau_h, \gamma_l))$ | Converges to unique pure NE |
| $D_1(0.3, 0.1)$ | $((\tau_l, \gamma_h), (\tau_l, \gamma_h))$ | Converges to unique pure NE |
| $D_2(0.01, 0.95, \frac{1}{2.7}, \frac{1}{2.6})$ | $NE_1'$: $((\tau_h, \gamma_l), (\tau_l, \gamma_h))$,<br>$NE_2'$: $((\tau_l, \gamma_h), (\tau_h, \gamma_l))$,<br>$NE_3'$: $((0, c', 1-c', 0), (0, c', 1-c', 0))$ | Depends on initialization:<br>Converges to the pure $NE_1'$ or $NE_2'$<br>or the mixed $NE_3'$ |
| $D_1(0.1, 0.2)$ | $NE_1$: $((\tau_l, \gamma_h), (\tau_l, \gamma_h))$,<br>$NE_2$: $((\tau_h, \gamma_l), (\tau_h, \gamma_l))$,<br>$NE_3$: $((0, c, 1-c, 0), (0, c, 1-c, 0))$ | Depends on initialization:<br>Converges to pure $NE_1$ or $NE_2$ |

Table 2: Nash Equilibria and Algorithm Dynamics for Different Distributions

Figures 2 to 7 represent experiments from full-information (left), where utility is calculated assuming demand from entire distribution of customers within threshold bounds, and sampling setting (right) where utility is calculated from a single sample of the distribution. No. of rounds $T = 50,000$, learning rate $\eta = 1/\sqrt{T}$.

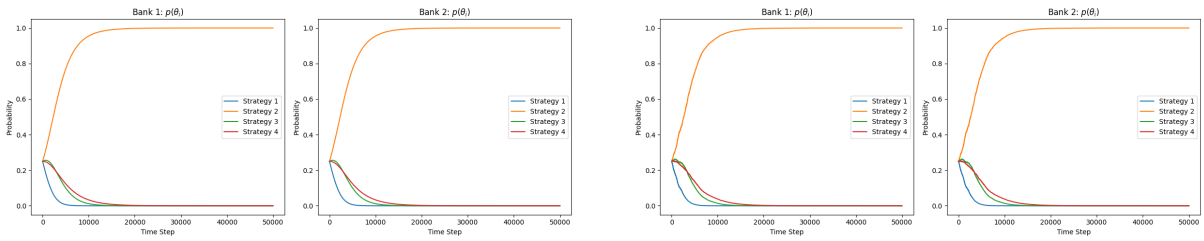


Figure 2: Convergence to pure Nash equilibrium with strategy 2 for both banks. Due to small variance, returns at higher credit scores is less, reducing utility for strategy 3.

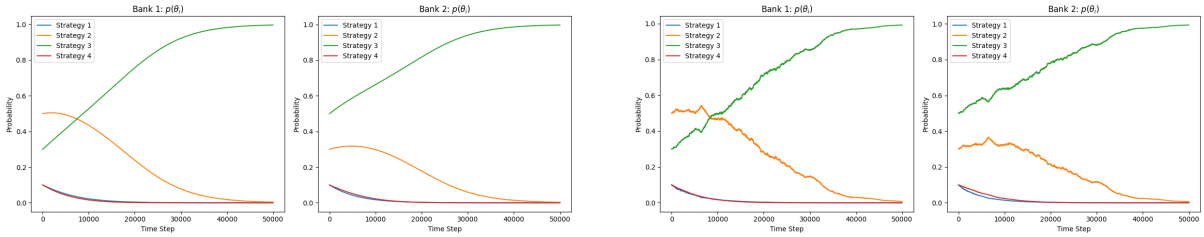


Figure 3: Convergence to pure Nash equilibrium with strategy 3 for both banks. Due to significant variance, returns at higher credit scores is significant, leading to slow convergence for strategy 3.

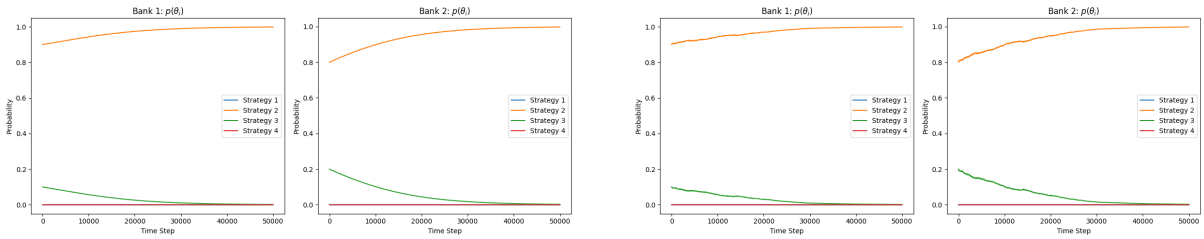


Figure 4: Convergence to pure Nash equilibrium with strategy 2 for both banks. Due to high initial probability for strategy 2, estimated returns for strategy 3 is less with both banks.

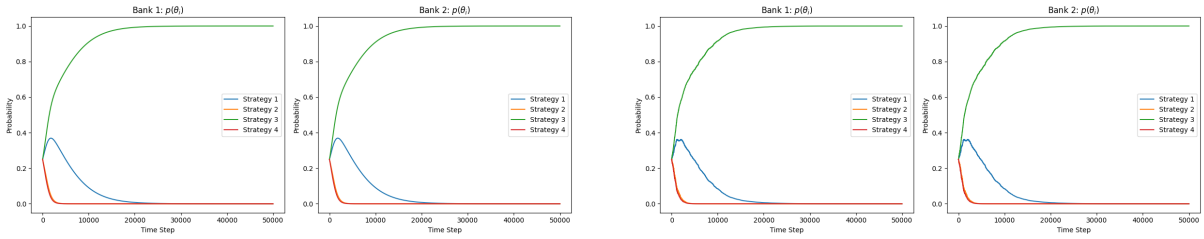


Figure 5: Convergence to pure Nash equilibrium with strategy 3 for both banks. Due to high variance, returns at higher credit scores is high, increasing utility for strategy 3.

# 6 Insights and Discussion

It is interesting to think about generalization of these results to an $n$-Bank Game with $n^2$ strategies, formulated as below.

- Consider n banks with parameters $\tau \in \{\tau_1, \ldots, \tau_n\}$, $\gamma \in \{\gamma_1, \ldots, \gamma_n\}$ (sorted in ascending), where $\tau_k = \frac{1}{2+\gamma_{n-k}}$.
- For bank $k$, customer distribution is

$$p_k(\theta_1, \ldots, \theta_n, y) = \begin{cases} 0 & \text{if } y < \tau_k \text{ or } \exists\, \tau_i \leq y (i \neq k) \text{ s.t. } \gamma_i < \gamma_k \\ \frac{p(y)}{Z_k(\theta_1,\ldots,\theta_n)\left(1+\sum_{i\neq k} \mathbb{I}[y_i=y_k]\mathbb{I}[y_j \geq \tau_j]\right)} & \text{otherwise} \end{cases}$$

(Normalization: $Z_k(\theta_1, \ldots, \theta_n)$)

This is also a non-Lipschitz distribution over varying $\theta$.

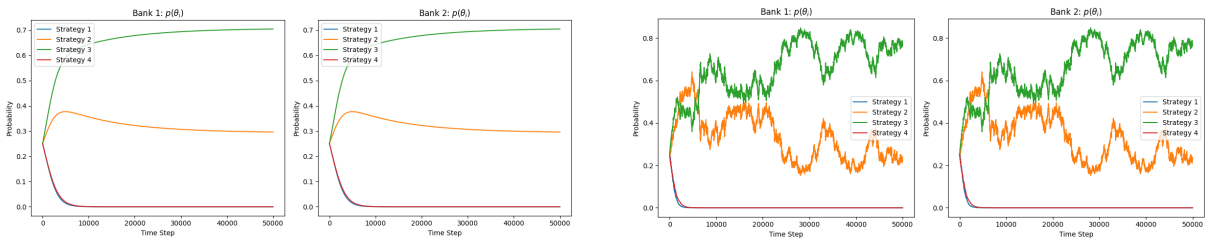


Figure 6: Convergence to mixed Nash equilibrium with strategy 2, 3 for both banks. Since utility is too low when both banks use same strategy in the piecewise uniform distribution, equal initial distribution induces mixed Nash equilibrium.

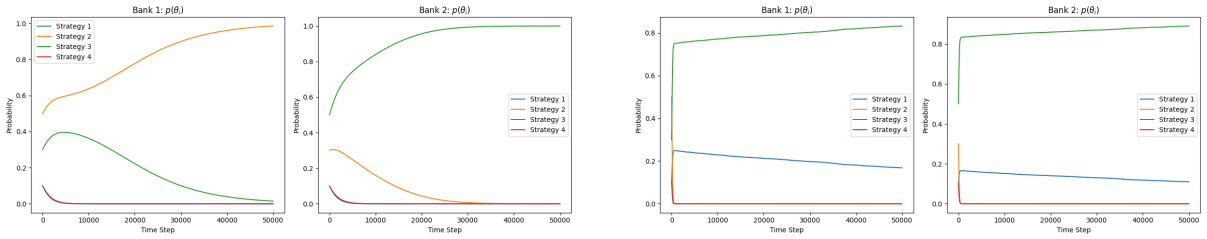


Figure 7: Convergence to asymmetric Nash equilibrium with strategy 2, 3 for both banks. Since utility is too low when both banks use same strategy in the piecewise uniform distribution, unequal initial distribution induces asymmetric Nash equilibrium. Single sample case doesn't converge due to extremely low variance distribution.

- The utility function for bank $k$ can be represented as $u_k(\theta_1, \ldots, \theta_n) = \mathbb{E}_{y \sim P_k}\left[(z + \gamma_k)g - 1\right]$
  Since number of banks is $n$ and number of strategies is $n^2$, $\mathcal{S}_{utility-matrix} = \mathcal{O}(n^{2n})$

For any bank, convergence of $n^2$ strategies to $n$ strategies: $\{(\tau_1, \gamma_1), \ldots, (\tau_n, \gamma_n)\}$ is guaranteed using Lemma 1 from the paper, since each strategy $(\tau_j, \gamma_k) \rightarrow (\tau_{n-k}, \gamma_k)$ since the delta utility $\epsilon > 0$. However, this reduces the search space from $O(n^{2n})$ to $O(n^n)$. Further analysis of convergence may require techniques such as vertex enumeration (as mentioned in the paper for 9-strategy case), and more time and effort to pursue such techniques.

Summing up, this paper discusses performative prediction, specifically in the context of Bank game, where prediction of parameters related to loan approval by a certain Bank effects the customer distribution. For instance, offering a certain parameter could shift the customer behavior due to certain members of population learning from their peers that underwent the Bank loan process. As a result a realistic assumption to make is the shift in the probability distribution of behavior of customers. The model discussed in the paper does not account for distribution shifts. Although in the incomplete information case, the Exponential Weights algorithm uses samples from distribution, the theoretical guarantees and the empirical evaluation assume/pick samples from the *same* distribution. Therefore, an interesting open question is to model probability distribution shifts based on adversarial behavior of agents (here customers). Although modeling shift in features is common place in Strategic learning/classification, we are not aware of how well studied distribution shifts are, although we found one very relevant work [2]. Given similarities of performative prediction literature with that of Strategic classification, it would also be interesting to see the use case of current Exponential Weights algorithm with sample information to be used in a long-time dynamical setting where agent changes their features upon observing the model parameters [4]. The above setting is also another use case to model distribution shifts. Another piece of discussion would be on dropping down the insensitivity assumption talked about in this paper. In the empirical evaluations, the choices of distributions used (piecewise uniform, bounded Gaussian) break the Lipshitz assumption. It would be interesting to establish general-purpose results based on insensitivity assumptions. Also, this paper specifically discusses two player game (or two Banks specifically) dynamics. An interesting obvious generalization of this framework would be to generalize to an $n$ player game which is not trivial. Another interesting avenue of exploration is to design initial conditions either based on prior experience to model realistic dynamics or help favor convergence outcomes. Throughout the work, acceleration of

convergence has not been much of a concern as theoretical and empirical evaluations show exponential convergence in all the cases, which comes from using Exponential algorithm. What came as surprise when implementing was that already stated in the paper, i.e we empirically observed a similar convergence rate when only one customer sample is available per time step.

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